FF–N–105a
July 26, 1963
SUPERSEDING
Fed. Spec. FF-N-105
October 6, 1952

FEDERAL SPECIFICATION

NAILS, WIRE, BRADS, AND STAPLES

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers wire nails, smooth and mechanical deformed shanks, and staples most suitable for general requirements, and special nails. Small nails and cut tacks are covered by Federal Specification FF-N-103.

1.2 Classification.

1.2.1 *Types.* Wire nails and staples shall be of the following types, styles, and sizes as specified (see 6.1):

Type I.—Brads:
Style 1.—Common (table I).

Type II.—Nails:
Style 1.—Asbestos-Board (table II).
2.—Barrel (table III).
3.—Boat (table IV).
4.—Box (table V).
5.—Broom (table VI).
6.—Casing (table VII).
7.—Cement-Coated Standard (Coolers) (table VIII).
8.—Cement-Coated Countersunk-Head (Sinkers) (table IX).
9.—Cement-Coated Countersunk Railroad (Corkers) (table X).
10.—Common (table XI).
11.—Concrete (table XII).
12.—Double Headed (table XIII).
13.—Fine (table XIV).
14.—Finishing (table XV).
15.—Flooring (table XVI).
16.—Lath (table XVII).
17.—Masonry (table XVIII).
18.—Pallet (table XIX).
19.—Plaster Board (table XX).
20.—Roofing (table XXI).
21.—Shingle (table XXII).
22.—Siding (table XXIII).
23.—Slating (table XXIV).
24.—Rubber Heel (table XXV).
25.—Underlayment (table XXVI).

Type III.—Staples:
Style 1.—Fence (table XXVII).
2.—Poultry Netting (table XXVIII).
3.—Flat Top Crown (table XXIX).
4.—Preformed (table XXX).

1.2.2 *Sizes.* Wire nails and staples shall be of the sizes listed hereinafter, and as specified in the invitation for bids. (See 6.1.)

1.2.3 *Finishes.* Wire nails and staples shall have the finishes as indicated hereinafter, and as specified in the invitation for bids. (See 6.1.)

1.2.4 *Shank design.* Wire nails shall be barbed, smooth or mechanically deformed, as indicated hereinafter, and as specified in the invitation for bids. (See 6.1.)

1.2.5 *Metals.* Wire nails and staples shall be made of the materials indicated herein-

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after, and as specified in the invitation for bids. (See 6.1.)

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitations for bids, or request for proposal form a part of this specification to the extent specified herein.

Federal Specifications:

NN-K-231—Kegs, Wood Slack.
TT-R-251—Remover; Paint (Organic Solvent Type).
PPP-B-566 — Boxes, Folding, Paperboard.
PPP-B-576—Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585—Boxes, Wood, Wirebound.
PPP-B-591—Boxes, Fiberboard, Wood Cleated.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed, and Lock-Corner.
PPP-B-636—Box, Fiberboard.
PPP-B-645—Boxes, Folding, Fiberboard, Heavy Duty.
PPP-B-665—Boxes, Paperboard, Metal-Stayed (Including Stay Material).
PPP-B-676—Boxes, Set-Up, Paperboard.
PPP-D-723—Drums, Fiber.
PPP-P-704 — Pails: Shipping, Steel (1 through 12 gallons).
PPP-T-97 — Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Test Method Std. No. 151—Metals: Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-L-10547—Liners, Case, Waterproof.

Military Standard:

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing and Materials Publication:

D 1037-60T—Wood-Base Fiber and Particle Panel Materials, Evaluating the Properties of.

C 380-63—Annular Ringed Nails for Gypsum Wallboard, Tentative Specifications for.

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race St., Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Materials.

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3.1.1 *Steel wire.* Steel wire used in the manufacture of nails and staples shall be of good commercial quality, entirely suitable for the purpose, and shall be sufficiently ductile to insure that the finished product will withstand without fracture cold bending through 180 degrees over a radius not greater than the diameter of the wire. The cold bend test will not be applied to barbed steel nails, or nails having mechanically deformed shanks. (See 4.6.1.)

3.1.2 *Hardened steel.* Hardened-steel nails shall have the minimum-carbon content of 1020 steel and shall be heat treated, quenched and tempered. When specified, hardened-steel nails when tested by any suitable means, shall be at least as stiff as low-carbon steel nails of the next larger gage. They shall bend at least 20 degrees before breaking. Nails with mechanically deformed shanks shall be compared with similar nails (See 4.6.1.) *WARNING: When using hardened nails, the use of safety goggles is recommended.*

3.1.3 *Copper.* Copper shall show on analysis not less than 98 percent pure copper. Copper nails shall withstand cold bending through 180 degrees over a radius not greater than the diameter of the nail, without fracture. (See 4.6.1.)

3.1.4 *Copper-clad wire.* Copper-clad wire used in the manufacture of staples shall be made by casting copper around a steel billet in such a manner as to obtain a firm weld between the two metals, or by any other process that will create an equal weld, and subsequently reducing to the desired size by hot rolling and cold drawing. Not less than 20 percent of the weight of the finished wire shall be copper. The average thickness of the copper shall be not less than 10 percent of the radius of the finished wire; the minimum thickness shall be not less than 8 percent of the radius of the finished wire. Staples produced from the wire shall withstand cold bending through 180 degrees over a radius not greater than the diameter of the shingles without fracture and without separation of the copper from the steel. (See 4.6.1.)

3.1.5 *Aluminum alloy.* Aluminum-alloy wire used in the manufacture of nails shall be of either of the following compositions, at the option of the contractor: 5056 alloy which shall be cold worked (5056-H193) and 6061 alloy heat treated and cold worked (6061-T913), to a minimum tensile strength of 63,000 lbs. per square inch. Smooth-shank nails shall be chemically treated to remove all grease, oil, and other foreign matter and to microscopically roughen the surface. Barbed and mechanically deformed shank nails shall be cleaned to remove all grease and other foreign matter. Finished smooth-shank nails shall withstand without fracture a cold bending through 90 degrees, over a radius not greater than the diameter of the nail. (See 4.6.1.)

Element	Percent			
	5056		6061	
	Min.	Max.	Min.	Max.
Magnesium	4.50	5.60	0.80	1.20
Silicon	—	.30	.40	.80
Copper	—	.10	.15	.40
Iron	—	.40	—	.70
Chromium	.05	.20	.15	.35
Zinc	—	.10	—	.25
Titanium	—	—	—	.15
Manganese	.05	.20	—	.15
Other elements, each	—	.05	—	.05
Other elements, total	—	.15	—	.15
Aluminum	Remainder		Remainder	

3.2 **Finishes.**

3.2.1 *Galvanized or zinc coating.* Galvanized or zinc-coated nails shall be reasonably smooth and shall be free from black and uncoated spots, fins, and lumps. The coating shall be not less than one ounce per square foot of surface area when tested in accordance with 4.6.7.

3.2.2 *Cement coating.* Cement coatings shall be reasonably even and shall not be

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tacky or gummy at normal temperatures or atmospheric conditions. Cement-coated nails having nominal diameters of 0.135 inch or less shall show an average increase in immediate holding power, above that of identical uncoated nails of not less than 50 percent when tested. (See 4.6.2.)

3.2.3 *Chemical etching*. This type nail shall be acid or chemically etched to produce a nail of increased holding power, and shall be as specified in the invitations for bids or proposals. This finish may be specified for any standard nail where no other finish is satisfactory for the requirement. Nails of all styles may be chemically etched when specified.

3.3 Mechanically deformed shanks. Mechanically deformed shanks may be of either the screw type or the annular-ring (complete circle) type.

3.4 Dimensions.

3.4.1 Nails with flat heads or with square shoulders under the heads shall be measured from under the head to the tip of the point and, except as otherwise specified, all other nails shall be measured overall; i.e., top of head to tip of point.

3.4.2 Brad-head, flat countersunk head, oval countersunk head, and cement-coated nails, except common box nails and standard cement-coated nails (coolers) shall be measured overall; i.e., top of head to end of point.

3.4.3 *Staples*.

3.4.3.1 *Fence and poultry netting staples*. These styles of staples shall be measured from the underside of the crown to the tip of the point for the leg lengths, and for the width shall be measured at the shoulder, inside of the legs.

3.4.3.2 *Power driven staples*. These staples are generally referred to by "crown width" and "leg length". The crown width is an approximate measurement outside of the legs, dependent on the gauge of the material. The inside measurement is critical and shall be as specified (see table XXIX and 6.1). The leg length is measured from the inside of the crown to the tip of the point.

3.4.4 Lengths are given hereinafter in inches and common fractions; diameters are given in common fractions when in multiples of one sixty-fourth of an inch, otherwise diameters are given in decimals of an inch.

3.4.5 Variation in the diameter of shanks of nails and staples from the dimensions specified shall not exceed ± 0.002" for diameters 0.076" or larger, and ± 0.001" for diameters smaller than 0.076". Measurement of diameters shall not include coatings except for copper-clad staples. Diameters of pointed nails shall be measured immediately below the gripper marks, variation in length shall not exceed ± 1/32" for nails and staples up to and including 1-1/2" in length and ± 1/16" for nails and staples over 1-1/2" in length. Diameters of mechanically deformed shanks shall be measured before deformation. The head diameter of roofing nails shall be plus 0 minus 10 percent of the nominal head diameter (mean of two readings 90 degree apart) and the head diameters for other nails shall be plus or minus 5 percent of the nominal head diameter (individual measurement). The difference in diameter across the long axis of a roofing nail head shall not exceed that across the short axis by more than 20 percent. For other nails the difference in diameter across the long axis should not exceed that across the short axis by more than 10 percent. A fillet must be provided under the head, an angle fillet is acceptable in lieu of a radius. Heads must be well centered.

3.4.6 In the following tables, the various dimensions and other data are indicated as follows:

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S—Nominal size or trade designation.
L—Nominal length in inches.
H—Nominal diameter or width of head in inches.
D—Nominal wire diameter in inches.
No/lb.—Approximate number per pound.

B—Bright or blued.
Z—Zinc-coated.
CM—Cement-coated.
MD—Mechanically deformed.
C—Crown width of staples.

Table I. Brads, type I, style 1, common. Brad head, diamond point, smooth, bright.



L.	D.	No./lb.	L.	D.	No./lb.	S.	L.	D.	No./lb.
Inches	*Inch*		*Inches*	*Inch*			*Inches*	*Inch*	
3/8	0.0348	9,520	1/2	0.0475	3,990	.....	3/4	0.0625	1,550
1/2	.0348	7,060	5/8	.0475	3,200	.....	7/8	.0625	1,280
5/8	.0348	5,680	3/4	.0475	2,620	2d	1	.072	876
3/4	.0348	4,800	7/8	.0475	2,220	3d	1-1/4	.080	568
7/8	.0348	4,220	1	.0625	1,120	4d	1-1/2	.099	316
1	.054	1,502	1-1/4	.0625	940	5d	1-3/4	.099	270
1-1/4	.054	1,208	1-1/2	.080	470	6d	2	.113	181
1-1/2	.054	1,040	1-3/4	.080	400	7d	2-1/4	.113	160
1-3/4	.0625	672	2	.080	350	8d	2-1/2	.131	107
2	.0625	580	.....	.....	.....	9d	2-3/4	.131	97
2-1/4	.080	320	.....	.....	.....	10d	3	.1483	70
2-1/2	.080	290	.....	.....	.....	12d	3-1/4	.1483	65
.....	.....	.....	.....	.....	.....	16d	3-1/2	.162	50
.....	.....	.....	.....	.....	.....	20d	4	.192	31
.....	.....	.....	.....	.....	.....	30d	4-1/2	.207	24
.....	.....	.....	.....	.....	.....	40d	5	.2253	18
.....	.....	.....	.....	.....	.....	50d	5-1/2	.2437	16
.....	.....	.....	.....	.....	.....	60d	6	.2625	11

Table II. Nails, type II, style 1, asbestos board. Aluminum alloy, flat head or casing head (for flush nailing).

L.	D.	Casing head		Flat head		Thickness of asbestos board
		H.	No./lb.	H.	No./lb.	
Inches	*Inch*	*Inch*		*Inch*		
1-1/8	0.099	9/64	1,046	5/16	988	1/8
1-1/4	.113	9/64	787	5/16	785	3/16
1-1/4	.135	5/32	516	7/16	491	1/4

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Table III. Nails, type II, style 2, barrel. Flat head, diamond point, bright or cement coated.



L.	D.	H.	No./lb. B.	No./lb. Cm.
Inches	*Inch*	*Inch*		
5/8	0.067	0.1483	1,615	1,570
3/4	.067	.1483	1,346	1,315
7/8	.076	.177	906	854
1	.076	.177	775	750
1-1/?	.076	.177	700	670
1-1/4	.080	.1875	568	539
1-3/8	.0915	.2187	400	386
1-1/2	.0915	.2187	367	355

Table IV. Nails, type II, style 3, boat. Oval countersunk head, chisel point, bright or zinc coated.



S.	L.	Light D.	Light H.	Light No./lb. B.	Light No./lb. Z.	Heavy D.	Heavy H.	Heavy No./lb. B.	Heavy No./lb. Z.
	Inches	*Inch*	*Inch*			*Inch*	*Inch*		
4d	1-1/2	3/16	13/32	82	79	1¼	1/2	44	43
6d	2	3/16	13/32	62	59	1/4	1/2	32	31
8d	2-1/2	3/16	13/32	50	48	1/4	1/2	26	25
10d	3	1/4	1/2	24	21	3/8	3/4	14	13
12d	3-1/4	1/4	1/2	22	19	3/8	3/4	13	13
16d	3-1/2	1/4	1/2	20	18	3/8	3/4	12	12
20d	4	1/4	1/2	18	16	3/8	3/4	10	10

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Table V. Nails, type II, style 4, box, miscellaneous. Flat head, cement coated, diamond point.



Box Nail, Apple, Cement-Coated



Egg-Case Nail, Cement-Coated



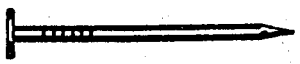
Box Nail, Fruit, Cement-Coated



Box Nail, Orange, Cement-Coated

Designation	S.	L.	D.	H.	No./lb.
		Inches	*Inch*	*Inch*	
Apple-box	5-1/2d	1-3/4	0.080	15/64	388
Egg-case	8d	1-1/8	.072	7/32	732
Fruit-box	4d	1-3/8	.072	7/32	599
Orange-box	4d	1-1/4	.072	7/32	654

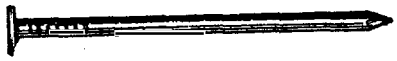
Table V-a. Nails, type II, style 4, box, veneer. Flat head, needle point.



S.	L.	D.	H.	No./lb.
	Inches	*Inch*	*Inch*	
4d	1-1/2	0.080	15/64	435

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Table V-b. Nails, type II, style 4, box, standard. Broad flat head, diamond point, barbed or smooth, bright or cement coated.



S.	L.	D.	H.	No./lb.
	Inches	*Inch*	*Inch*	
2d	1	0.067	3/16	1,010
3d	1-1/4	.076	7/32	635
4d	1-1/2	.080	7/32	473
5d	1-3/4	.080	7/32	406
6d	2	.099	17/64	236
7d	2-1/4	.099	17/64	210
8d	2-1/2	.113	19/64	145
9d	2-3/4	.113	19/64	132
10d	3	.128	5/16	94
12d	3-1/4	.128	5/16	88
16d	3-1/2	.135	11/32	71
20d	4	.1483	3/8	52
30d	4-1/2	.1483	3/8	46
40d	5	.162	13/32	35

Table V-c. Nails, type II, style 4, box, standard. Flat head, diamond point, cement coated.



S.	L.	D.	H.	No./lb.
	Inches	*Inch*	*Inch*	
2d	1	0.058	11/64	1,252
3d	1-1/8	.0625	3/16	978
4d	1-3/8	.067	13/64	680
5d	1-5/8	.072	7/32	510
6d	1-7/8	.086	1/4	315
7d	2-1/8	.086	1/4	280
8d	2-3/8	.099	17/64	190
9d	2-5/8	.099	17/64	172
10d	2-7/8	.113	19/64	119

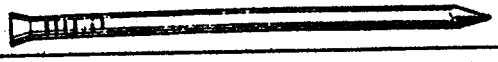
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Table VI. Nails, type II, style 5, broom. Flat or star head, diamond point.



L.	D.	H.	No./lb.
Inches	*Inch*	*Inch*	
5/8	0.072	13/64	1,350
5/8	.080	7/32	1,015
3/4	.072	13/64	1,090
3/4	.080	7/32	855

Table VII. Nails, type II, style 6, casing. Deep countersink head, diamond point.



S.	L.	D.	H.	No./lb.
	Inches	*Inch*	*Inch*	
2d	1	0.067	0.099	1,090
3d	1-1/4	.076	.113	654
4d	1-1/2	.080	.1205	489
5d	1-3/4	.080	.1205	414
6d	2	.099	.142	244
7d	2-1/4	.099	.142	215
8d	2-1/2	.113	.155	147
9d	2-3/4	.113	.155	133
10d	3	.128	.170	96
12d	3-1/4	.128	.170	88
16d	3-1/2	.135	.177	74

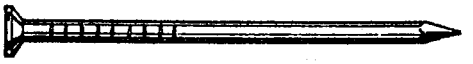
Table VIII. Nails, type II, style 7, cement coated, standard (coolers) Flat head, diamond point.



S.	L.	D.	H.	No./lb.
	Inches	*Inch*	*Inch*	
2d	1	0.0625	11/64	1,110
3d	1-1/8	.067	3/16	839
4d	1-3/8	.080	7/32	493
5d	1-5/8	.086	15/64	366
6d	1-7/8	.0915	1/4	278
7d	2-1/8	.099	17/64	212
8d	2-3/8	.113	9/32	144
9d	2-5/8	.113	9/32	131
10d	2-7/8	.1205	19/64	105

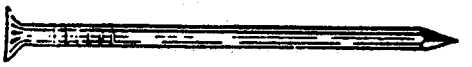
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Table IX. Nails, type II, style 8, cement coated (sinkers). Flat countersunk head, diamond point.



S.	L.	D.	H.	No./lb.
	Inches	*Inch*	*Inch*	
3d	1-1/8	0.067	11/64	923
4d	1-3/8	.080	13/64	527
5d	1-5/8	.086	7/32	387
6d	1-7/8	.0915	15/64	293
7d	2-1/8	.099	1/4	223
8d	2-3/8	.113	17/64	153
10d	2-7/8	.1205	9/32	111
12d	3-1/8	.135	5/16	81
16d	3-1/4	.1483	11/32	64
20d	3-3/4	.177	3/8	40
30d	4-1/4	.192	13/32	30
40d	4-3/4	.207	7/16	23
60d	5-3/4	.2437	1/2	14

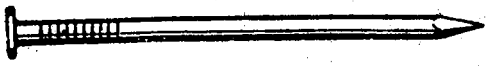
Table X. Nails, type II, style 9, cement coated, railroad (corkers). Flat countersunk head, diamond point.



S.	L.	D.	H.	No./lb.
	Inches	*Inch*	*Inch*	
2d	1	0.0625	5/32	1,217
3d	1-1/4	.072	3/16	720
4d	1-1/2	.086	7/32	419
5d	1-5/8	.086	7/32	387
6d	1-7/8	.099	1/4	253
7d	2-1/8	.099	1/4	223
8d	2-3/8	.1205	9/32	135
9d	2-5/8	.1205	9/32	122
10d	2-7/8	.135	5/16	89
12d	3-1/8	.135	5/16	81
16d	3-3/8	.1483	11/32	62
20d	3-7/8	.177	3/8	38
30d	4-3/8	.192	13/32	29
40d	4-7/8	.207	7/16	22
50d	5-3/8	.2253	15/32	17
60d	5-7/8	.2437	1/2	13

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Table XI. Nails, type II, style 10, common, aluminum-alloy. Flat head, diamond point.



S	L	D	H	No./lb.
	Inches	Inch	Inch	
4d	1-1/2	.099	1/4	825
6d	2	.120	17/64	430
8d	2-1/2	.148	9/32	220
10d	3	.162	5/16	170
16d	3-1/2	.177	11/32	120
20d	4	.199	13/32	78

Table XI-a. Nails, type II, style 10, common, solid copper. Flat head, diamond point.



L.	D.	No./lb.	L.	D.	No./lb.
Inches	*Inch*		*Inches*	*Inch*	
5/8	0.065	1,380	2	0.120	180
3/4	.065	1,160	2	.134	106
3/4	.072	956	2-1/2	.134	86
7/8	.072	808	3	.148	56
1	.072	704	3-1/2	.165	40
1-1/4	.083	424	4	.203	23
1-1/2	.109	208	4-1/2	.220	18
1-3/4	.109	180	5	.238	14
1-3/4	.120	144	6	.284	8

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Table XI-b. Nails, type II, style 10, common, steel. Flat head, diamond point, bright, zinc or cement coated, smooth.

S.	L.	D.	H.	No./lb. Z.	No./lb. Other
	Inches	*Inch*	*Inch*		
2d	1	0.072	11/64	789	810
3d	1-1/4	.080	13/64	506	540
4d	1-1/2	.099	1/4	274	300
5d	1-3/4	.099	1/4	236	255
6d	2	.113	17/64	155	170
7d	2-1/4	.113	17/64	140	150
8d	2-1/2	.131	9/32	91	95
9d	2-3/4	.131	9/32	86	90
10d	3	.1483	5/16	61	65
12d	3-1/4	.1483	5/16	57	60
16d	3-1/2	.162	11/32	44	46
20d	4	.192	13/32	26	28
30d	4-1/2	.207	7/16	20	20
40d	5	.2253	15/32	17	17
50d	5-1/2	.2437	1/2	13	13
60d	6	.2625	17/32	10	10



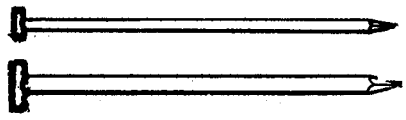
Table XI-c. Nails, type II, style 10, common, regular, stiff stock or hardened steel, flat head, diamond point, mechanically deformed.



Size	Length	Gauge	Wire Diameter	Head	No./lb
6d	2"	11	.120	9/32"	147
8d	2-1/2"	11	.120	9/32"	120
10d	3"	10	.135	5/16"	78
12d	3-1/4"	10	.135	5/16"	73
16d	3-1/2"	9	.148	5/16"	57
20d	4"	7	.177	3/8"	35
40d	5"	7	.177	3/8"	27
60d	6"	7	.177	3/8"	23
70d	7"	5	.207	1/2"	15
80d	8"	5	.207	1/2"	13

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Table XI-d. Nails, type II, style 10, common, T-head, diamond point, aluminum-alloy, bright or galvanized steel. (Power driven.)



Length	5/32 In. Head Length Dia	17/64 In. Head Length Dia
1 1/2	.080 .086	.080 .086 .097
1 5/8	.080 .086	.080 .086 .097
1 3/4	.080 .086	.080 .086 .097
1 7/8	.080 .086	.080 .086 .097
2	.080 .086	.080 .086 .097

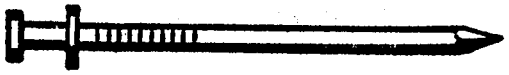
Table XII. Nails, type II, style 11, concrete. Flat countersunk head, diamond point, hardened.



L.	D.	H.
Inches	*Inch*	*Inch*
1/2	0.1483	5/16
5/8	.1483	5/16
3/4	.1483	5/16
7/8	.1483	5/16
1	.1483	5/16

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Table XIII. Nails, type II, style 12, double-headed. Diamond point.



S.	L.[a]	D.	Distance between heads	No./lb.
	Inches	*Inch*	*Inches*	
6d	1-3/4	0.113	1/4	156
8d	2-1/4	.131	1/4	90
10d	2-3/4	.1483	5/16	59
16d	3	.162	3/8	45
20d	3-1/2	.192	3/8	28
30d	4	.207	7/16	22

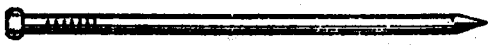
L-Length from end of point to underside of lower head.

Table XIV. Nails, type II, style 13, fine. Flat head. diamond point.



S.	L.	D.	H.	No./lb.
	Inches	*Inch*	*Inch*	
8d	1-1/8	0.072	11/64	757

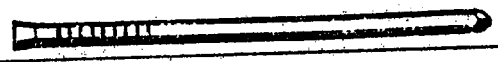
Table XV. Nails, type II, style 14, finishing. Brad head, diamond point.



S.	L.	D.	H.	No./lb.
	Inches	*Inch*	*Inch*	
2d	1	0.058	0.086	1,473
3d	1-1/4	.067	.099	880
4d	1-1/2	.072	.1055	630
5d	1-3/4	.072	.1055	535
6d	2	.0915	.135	288
7d	2-1/4	.0915	.135	254
8d	2-1/2	.099	.142	196
9d	2-3/4	.099	.142	178
10d	3	.113	.155	124
12d	3-1/4	.113	.155	113
16d	3-1/2	.1205	.162	93
20d	4	.135	.177	65

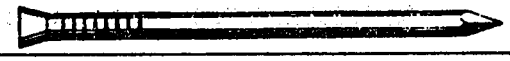
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Table XVI. Nails, type II, style 15, flooring. Countersunk head, blunt point, bright or cement coated.



S.	L.	D.	H.	No./lb. B.	No./lb. Cm.
8d	2-1/2	0.113	0.155	142	141

Table XVI-a. Nails, type II, style 15, flooring. Countersunk or casing head, diamond point, smooth or mechanically deformed shank, regular, stiff stock or hardened steel.



S.	L.	D.	H.	No./lb.
8d	2-1/2	0.135	0.177	105



Size	Length	Gauge	Dia	Head	No./lb
2d	1"	15	.072	9/64"	840
3d	1 1/4"	15	.072	9/64"	669
4d	1 1/2"	13	.091	5/32"	369
4d	1 1/2"	14	.083	5/32"	426
5d	1 3/4"	13	.091	5/32"	313
6d	2"	11 1/2	.115	13/64"	173
7d	2 1/4"	11 1/2	.115	13/64"	147
8d	2 1/2"	11 1/2	.115	13/64"	142
10d	3"	10	.135	1/4"	82
12d	3 1/4"	10	.135	1/4"	75
16d	3 1/2"	9	.148	9/32"	58

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Table XVII. Nails, type II, style 16, lath, wood. Flat head, diamond point, blued.



S.	L.	D.	H.	No./lb.
	Inches	*Inch*	*Inch*	
2d	1	0.058	9/64	1,280
3d (light)	1-1/8	.0625	5/32	984
3d	1-1/8	.072	11/64	757

Table XVII-a. Nails, type II, style 16, lath, metal. Thin, flat hook-head, diamond point, bright blued, or zinc coated.



L.	D.	H., length	No./lb. B.	No./lb. Z.
Inches	*Inch*	*Inch*		
1-1/8	0.1055	7/16	278	263

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Table XVIII. Nails, type II, style 17, masonry. Flat head, diamond point, mechanically deformed.



Length	Gauge	Dia	Head	No./lb
1/2	9	.148	5/16	339
3/4	9	.148	5/16	280
1	9	.148	5/16	173
1 1/4	9	.148	5/16	145
1 1/2	9	.148	5/16	129
1 3/4	9	.148	5/16	113
2	9	.148	5/16	98
2 1/4	9	.148	5/16	84
2 1/2	9	.148	5/16	76
2 3/4	9	.148	5/16	70
3	9	.148	5/16	67
3 1/4	9	.148	5/16	60
3 1/2	8	.162	11/32	48
3 3/4	8	.162	11/32	45
4	7	.177	3/8	35



Length	Gauge	Dia	Head	No./lb
1	-	.250	9/16	63
1 1/4	-	.250	9/16	47
1 1/2	-	.250	9/16	43
1 3/4	-	.250	9/16	39
2	-	.250	9/16	34
2 1/2	-	.250	9/16	27
3	-	.250	9/16	24
3 1/2	-	.250	9/16	19

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Table XIX. Nails, type II, style 18, pallet. Flat head, diamond point, mechanically deformed.

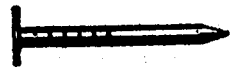


Size	Length	Gauge	Dia	Head	No./lb
-	1 1/2"	11	.120	9/32"	191
-	1 5/8"	11	.120	9/32"	170
-	2"	11	.120	9/32"	147
-	2 1/4"	11	.120	9/32"	131
-	2 1/2"	11	.120	9/32"	120
-	2 1/2"	10	.135	5/16"	95
-	3"	11	.120	9/32"	94
-	3"	10	.135	5/16"	78
-	3"	9	.148	5/16"	66
-	3 1/4"	9	.148	5/16"	60
-	3 1/4"	10	.135	5/16"	73
-	3 1/2"	9	.148	5/16"	57
-	3 1/2"	8	.162	3/8"	48
-	3 1/2"	7	.177	7/16"	40
-	4"	7	.177	7/16"	35

NOTE: The shank of the nails shall have helical threads extending up from the point at least 2/3 fo the shank length. Angulation of helical thread to be such that the projection of the thread across the axis of the nail shall make an angle of 60° ± 5° with a plane perpendicular to the axis. The point of the nail shall be diamond shaped, 5/32" long ± 1/16".

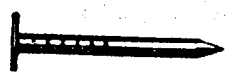
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Table XX. Nails, type II, style 19, plasterboard. Aluminum-alloy, flat head, diamond point.



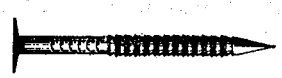
L.	D.	H.	No./lb.
Inches	*Inch*	*Inch*	
1-1/8	0.099	5/16	988
1-1/4	.099	5/16	939
1-1/2	.1055	5/16	725

Table XX-a. Nails, type II, style 19, plasterboard. Steel, flat head, diamond point, blued.



L.	D.	H.	No./lb.	L.	D.	H.	No./lb.
Inches	*Inch*	*Inch*		*Inches*	*Inch*	*Inch*	
1-1/8	0.0915	19/64	478	1-1/8	0.0915	3/8	449
1-1/4	.0915	19/64	425	1-1/4	.1055	3/8	310
				1-3/4	.0915	3/8	294

Table XX-b. Nails, type II, style 19, plasterboard. Steel, flat head, diamond point, mechanically deformed shank.



Length	Diameter	Head	No./lb
*1-1/8"	.098	1/4"	380
*1-1/4"	.098	1/4"	345
*1-3/8"	.098	1/4"	320
1-1/2"	.098	1/4"	290
1-5/8"	.098	1/4"	270

*Recommended by ASTM C 380-63

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Table XXI. Nails, type II, style 20, roofing, aluminum-alloy. Flat head, diamond point (for prepared felt roofing).



L.	D.	H.
Inches	*Inch*	*Inch*
1	0.135	9/16

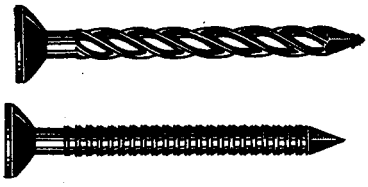
Table XXI-a. Nails, type II, style 20, roofing, aluminum-alloy. large head, diamond point.



		Approximate number per pound							
D.	H.	L.3/4	L.7/8	L.1	L.1-1/4	L.1-1/2	L.1-3/4	L.2	L.2-1/2
Inch	*Inch*								
0.120	7/16	936	832	759	616	523	.....	.....	.....
.135	7/16	746	663	605	491	417	368	336	.....
145	7/16	.....	.....	.....	.....	.....	.....	.....	227

Table XXI-b. Nails, type II, style 20, roofing, aluminum-alloy. Large head, diamond point, with neoprene washer attached (for aluminum-roofing sheet), smooth or mechanically deformed.

	Smooth shank			Screw shank		
L.	D.	H.	No./lb.	D.	H.	No./lb
Inches	*Inch*	*Inch*		*Inch*	*Inch*	
1-3/4	0.135	7/16	318	0.145	7/16	290
2	.135	7/16	285	.145	7/16	259
2-1/4	.135	7/16	243	.145	7/16	235
2-1/2	.145	7/16	208	.145	7/16	215



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Table XXI-c. Nails, type II, style 20, roofing, steel, checkered head, smooth, diamond point, bright or zinc coated.



L.	D.	H.	No./lb. B.	No./lb. Z.
Inches	*Inch*	*Inch*		
7/8	0.1205	1/2	242	227
1	.1205	1/2	220	204
1-1/4	.1205	1/2	186	166
1-1/2	.1205	1/2	160	151
1-3/4	.1205	1/2	141	125

Table XXI-d. Nails, type II, style 20, roofing, steel. Large head, barbed, diamond point, bright or zinc coated.



D.	H.	Approximate number per pound: L.3/4 in. B.	L.3/4 in. Z.	L.7/8 in. B.	L.7/8 in. Z.	L.1 in. B.	L.1 in. Z.
Inch	*Inch*						
0.1055	3/8	513	449	443	388	390	341
.1205	7/16	368	338	320	294	284	261
.135	15/32	310	266	269	231	236	203
.142	31/64	263	233	230	205	204	182
.1483	1/2	242	227	211	198	187	175
.162	1/2	219	206	188	177	166	156

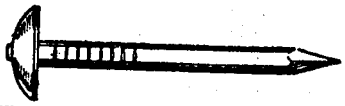
D.	H.	L.1-1/8 in. B.	L.1-1/8 in. Z.	L.1-1/4 in. B.	L.1-1/4 in. Z.	L.1-1/2 in. B.	L.1-1/2 in. Z.	L.1-3/4 in. B.	L.1-3/4 in. Z.
Inch	*Inch*								
0.1055	3/8	348	304	314	274	263	230	226	198
.1205	7/16	255	234	231	212	194	178	169	155
.135	7/16	211	181	191	164	162	139	137	118
142	31/64	183	162	165	146	139	123	121	107
.1483	1/2	167	156	152	142	128	120	111	104
.162	1/2	148	139	133	125	112	106	96	90

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Table XXI-e. Nails, type II, style 20, roofing, steel. Large head, copper clad.

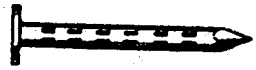
S.	L.	D.	H.	No./lb.
	Inches	*Inch*	*Inch*	
2d	1	0.120	3/8	280
3d	1-1/4	.120	3/8	220
4d	1-1/2	.120	3/8	190
5d	1-3/4	.120	3/8	160
6d	2	.120	3/8	140
7d	2-1/4	.120	3/8	130

Table XXI-f. Nails, type II, style 20, roofing, steel. Leak resistant, convex head, diamond point, zinc coated.



L.	D.	H.	No./lb.
Inches	*Inch*	*Inch*	
1-3/4	0.135	1/2	110
2	.135	1/2	100

Table XXI-g. Nails, type II, style 20, roofing, steel. Regular head, flat head, barbed, diamond point, zinc coated.



L.	D.	H.	No./lb.	
			B.	Z.
Inches	*Inch*	*Inch*		
1	0.1055	9/32	411	387
1-1/4	.1205	5/16	251	230

Table XXI-h. Nails, type II, style 20, roofing, steel, flat reinforced head. For prepared felt roofing. needle or diamond point, bright or zinc coated.

L.	H.	Approximate number per pound									
		L.3/4 in.		L.7/8 in.		L.1 in.		L.1-1/8 in.		L.1-1/4 in.	
		B.	Z.	B.	Z.	B.	Z.	B.	Z.	B.	Z.
Inch	*Inch*										
0.1055	5/8	210	180	195	172	180	160	170	152	176	148
.1205	5/8	180	164	175	157	162	145	149	135	136	---

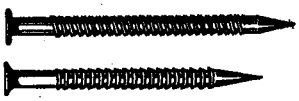
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Table XXI-i. Nails, type II, style 20, roofing, steel. One-inch integral steel cap, mechanically deformed shanks (for roofing felts).

Length	Gauge	Diameter	Head	No./lb
1/2"	12	.109	1"	128
5/8"	12	.109	1"	124
3/4"	12	.109	1"	115
7/8"	12	.109	1"	112
1"	12	.109	1"	110
1-1/8"	12	.109	1"	106
1-1/4"	12	.109	1"	104
1-1/2"	11 or 12	.120 or .109	1"	84 or 96
1-3/4"	11 or 12	.120 or .109	1"	78 or 94
2"	11 or 12	.120 or .109	1"	74 or 90
2-1/2"	11 or 12	.120 or .109	1"	61 or 80
3"	12	.109	1"	71

Table XXII. Nails, type II, style 21, shingle, aluminum-alloy, asbestos shingle. Flat head, diamond point, smooth or mechanically deformed.





L	D	H	No/lb	L	D	H	Nr./lb
Inches	Inch	Inch		1 1/4	.101	.191	1057
				1 1/2	.101	.191	856
1-1/4	0.113	5/16	785	1 3/4	.105	.191	722
1-1/2	.113	5/16	659	2	.105	.191	609
1-3/4	.113	5/16	644	2 1/4	.112	.200	477
				2 1/2	.112	.200	427

Table XXII-a. Nails, type II, style 21, shingle, aluminum-alloy, cedar shingle. Flat head, diamond point, smooth.

S.	L.	D.	H.	No./lb.
	Inches	*Inch*	*Inch*	
3d	1-1/4	0.080	7/32	1,480

Table XXII-b. Nails, type II, style 21, shingle, aluminum-alloy, cedar shake. Flat head, diamond point, smooth.

L.	D.	H.	No./lb.
Inches	*Inch*	*Inch*	
1-1/4	0.086	5/32	1,300
1-3/4	.099	5/32	724

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Table XXII-c. Nails, type II, style 21, shingle, aluminum-alloy, general purpose. Flat head, diamond point, smooth.

S.	L.	D.	H.	No./lb.
	Inches	*Inch*	*Inch*	
---	7/8	0.099	9/32	1,313
3d	1-1/4	.099	9/32	1,009

Table XXII-d. Nails, type II, style 21, shingle, steel. Shingle-asbestos, large flat head, diamond point, bright or zinc coated, barbed.



L.	D.	H.	No./lb. B.	No./lb. Z.
Inches	*Inch*	*Inch*		
1-1/4	0.113	13/32	261	238
1-1/2	.113	13/32	219	197
1-3/4	.113	13/32	190	170
2	.113	13/32	170	148

Table XXII-e. Nails, type II, style 21, shingle, steel, standard. Flat head, diamond point, smooth, bright or zinc coated.



S.	L.	D.	H.	No./lb. B.	No./lb. Z.
	Inches	*Inch*	*Inch*		
3d	1-1/4	0.092	1/4	434	400
3-1/2d	1-3/8	.099	9/32	331	306
4d	1-1/2	.1055	9/32	271	252

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Table XXIII. Nails, type II, style 22, siding, aluminum-alloy. Flat head, diamond point, smooth.



L.	D.	H.	No./lb.
Inches	*Inch*	*Inch*	
1-1/2	0.113	7/32	701
1-1/2	.113	5/16	659

Table XXIII-a. Nails, type II, style 22, insulated siding, aluminum alloy, Flat head, diamond point, smooth.



L.	D.	H.	No./lb.
Inches	*Inch*	*Inch*	
2	0.113	7/32	495
2-1/2	.135	7/32	295

Table XXIII-b. Nails, type II, style 22, wood siding, aluminum-alloy. Casing or countersunk head, diamond point, smooth.



S.	L.	D.	H. Casing head	H. Countersunk head	No./lb.
	Inches	*Inch*			
6d	1-7/8	0.1055	9/64	17/64	604
7d	2-1/8	.113	9/64	17/64	468
8d	2-3/8	.128	5/32	19/64	319
10d	2-7/8	.1483	3/16	5/16	197

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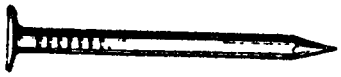


Table XXIV. Nails, type II, style 23, slating, aluminum-alloy. Large flat head, 5/16 to 3/8 inch, diamond point, smooth.

L.	D.	No./lb.
Inches	*Inch*	
7/8	0.1055	1,170
1	.1055	1,050
1-1/4	.1055	867
1-1/4	.1205	618
1-1/4	.135	519
1-1/2	.1205	528
1-1/2	.135	435

Table XXIV-a. Nails, type II, style 23, slating, solid copper. Flat head, diamond point.

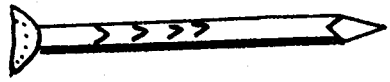
L.	D.	No./lb.
Inches	*Inch*	
1	0.109	292
1-1/4	.135	164
1-1/4	.120	208
1-1/4	.109	240
1-1/2	.134	132
1-1/2	.120	164
1-1/2	.109	204
1-3/4	.134	116
2	.134	108

Table XXIV-b. Nails, type II, style 23, slating, steel. Flat, slightly countersunk head, diamond point, bright.

S.	L.	D.	H.	No./lb.
	Inches	*Inch*	*Inch*	
2d	1	0.1055	5/16	411
3d	1-1/4	.128	3/8	225
4d	1-1/2	.128	3/8	187
5d	1-3/4	.135	13/32	142
6d	2	.1483	7/16	103

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Table XXV. Nails, type II, style 24, rubber heel. Flat or countersunk head, pencil or clinch point, bright.



Countersunk head and pencil point



large flat head and clinch point

L	D	H
5/8	0.08	0.154
3/4	0.08	0.154
7/8	0.08	0.154
1	0.08	0.154
1 1/8	0.08	0.154
1 1/4	0.08	0.154

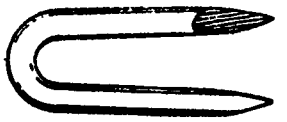
Table XXVI. Nails, type II, style 25, underlayment. Flat head, diamond point, steel, mechanically deformed.



Size	Length	Gauge	Dia	Head	No./lb
-	1"	14	.083	3/16"	624
-	1 1/4	14	.083	3/16"	513
-	1 1/4	12 1/2	.098	1/4"	362
-	1 3/8	14	.083	3/16"	445
-	1 3/8"	12 1/2	.098	1/4"	315
-	1 1/2"	14	.083	3/16"	409
-	1 1/2"	12 1/2	.098	1/4"	289
-	1 5/8"	12 1/2	.098	1/4"	278
-	1 3/4"	12 1/2	.098	1/4"	252
6d	1 7/8"	12	.105	17/64"	175
7d	2 1/8"	-	.109	17/64	171
8d	2 3/8	11 1/2	.115	19/64	141

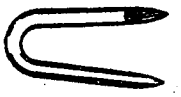
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Table XXVII. Staples, type III, style 1, fence. Bright and zinc coated.



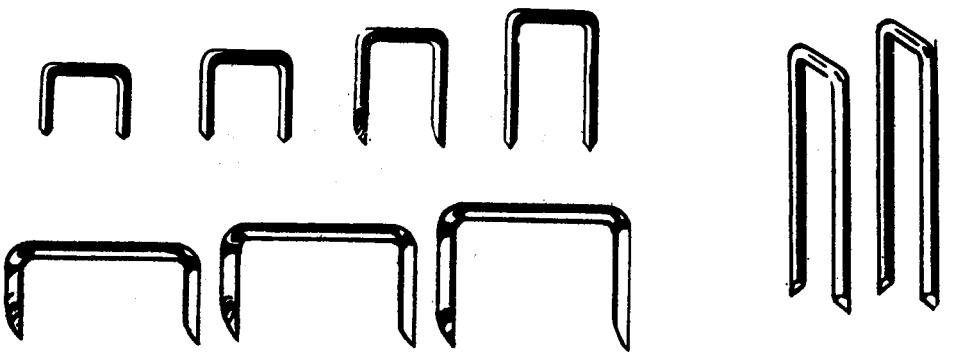
L. Bright	L. Zinc-coated	D.	No./lb. B.	No./lb. Z.
Inches	*Inches*	*Inch*		
7/8	.....	0.1483	122	.....
1	1	.1483	106	105
1-1/8	1-1/8	.1483	97	93
1-1/4	1-1/4	.1483	87	84
1-1/2	1-1/2	.1483	72	70
1-3/4	.....	.1483	61	.....

Table XXVIII. Staples, type III, style 2, poultry netting, zinc coated.



L.	D.	No./lb.
Inches	*Inch*	
3/4	0.080	499

Table XXIX. Staples, type III, style 3, flat top crown. Steel, aluminum or bronze, galvanized, cement coated or etched. (For use in pneumatic or mechanical guns in fastening wood.)



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Table XXIX. Continued.

3/16 Inch Crown Width Staples, 18 Gauge

Length	Inches Between Legs
3/8	0.097
1/2	0.097
5/8	0.097
3/4	0.097
7/8	0.097
1	0.097
1 1/8	0.097

1/2 Inch Crown Width Staples 14 and 16 Gauge

Length	Inches Between Legs	
	14 Gauge	16 Gauge
1/2	.340	-
5/8	.340	-
3/4	.340	-
7/8	.340	.375
1	.340	.375
1 1/8	.340	.375
1 1/4	.340	.375
1 3/8	.340	.375
1 1/2	.340	.375
1 5/8	.340	.375
1 3/4	.340	.375
1 7/8	.340	-
2	.340	-

7/16 Inch Crown Width Staples, 14, 15 and 16 Gauge.

Length	Incnes Between Leg		
	14 Gauge	15 Gauge	16 Gauge
3/8	-	-	.352
1/2	-	-	.352
5/8	-	-	.352
3/4	-	-	.352
7/8	.352	-	.352
1	.352	-	.352
1 1/8	.352	.352	.352
1 1/4	.352	.352	.352
1 3/8	.352	.352	.352
1 1/2	.352	.352	.352
1 5/8	.352	-	.352
1 3/4	.352	-	.352
1 7/8	.352	-	.352
2	.352	-	.352

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Table XXIX continued.

3/4 Inch Crown Width Staples, 14 and 16 Gauge.

Length	Inches Between Legs 14 Gauge	16 Gauge
1/2	.590	.625
5/8	.590	.625
3/4	.590	.625
7/8	.590	.625
1	.590	.625
1 1/8	.590	.625
1 1/4	.590	.625
1 3/8	.590	.625
1 1/2	.590	.625
1 5/8	.590	.625
1 3/4	.590	.625
1 7/8	.590	.625
2	.590	.625

1 Inch Crown Width Staples, 14 and 16 Gauge.

Length	Inches Between Legs 14 Gauge	16 Gauge
1/2	.900	.900
5/8	.900	.900
3/4	.900	.900
7/8	.900	.900
1	.900	.900
1 1/8	.900	.900
1 1/4	.900	.900

Table XXX. Staples, type III, style 4, preformed. Steel, cement-coated, galvanized, coppered, wash-tinned, or other plated finish. (To be hand driven.)

Staples, Common Double Pointed.

No.	Width	Length	Gauge	Flatten To	No./lb	Point Length	Point Angle
5	7/32	3/8	.054	.040	1920	3/16	13°
8	3/16	13/32	.068	.048	1376	3/16	12°
9	7/32	7/16	.068	.048	1248	1/4	12°
10	1/4	1/2	.072	.057	864	1/4	12°
11	9/32	9/16	.072	.057	800	5/16	12°
12	5/16	5/8	.072	.057	672	5/16	12°
14	5/16	11/16	.083	.060	544	11/32	12°
16	3/8	3/4	.083	.060	410	11/32	12°

All inside dimensions.

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Table XXX. Continued

Staples, Electricians

No.	Width	Length	Gauge	Flatten To	No./lb	Point Length	Point Angle
3/8	5/32	3/8	.068	.048	1440	1/4	12°
4/8	5/32	1/2	.072	.057	992	1/4	12°
5/8	7/32	5/8	.072	.057	736	5/16	12°
6/8	7/32	3/4	.083	.060	480	11/32	12°
7/8	7/32	7/8	.083	.060	416	11/32	12°
1"	7/32	1"	.083	.060	352	11/32	12°

All inside dimensions.

Staples, Hoop

1/2 Inch Wide Hoop Staples (Inside Measurement)

Length	Gauge	Flatten To	No./lb
1/2	.072	.057	720
1/2	.083	.060	467
5/8	.072	.057	576
5/8	.083	.060	432
3/4	.072	.057	494
3/4	.083	.060	375

5/8 Inch Wide Hoop Staples (Inside Measurement)

Length	Gauge	Flatten To	No./lb
1/2	.072	.057	628
1/2	.083	.060	471
5/8	.072	.057	532
5/8	.083	.060	399
3/4	.072	.057	461
3/4	.083	.060	340

3/4 Inch Wide Hoop Staples (Inside Measurement)

Length	Gauge	Flatten To	No./lb
1/2	.072	.057	576
1/2	.083	.060	432
1/2	.109	.083	256
5/8	.072	.057	493
5/8	.083	.060	370
5/8	.109	.083	219
3/4	.072	.057	432
3/4	.083	.060	324
3/4	.109	.083	192
1"	.072	.057	346
1"	.083	.060	259
1"	.109	.083	154

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Table XXX. Continued.

7/8 Inch Wide Hoop Staples (Inside Measurment)

Length	Gauge	Flatten To	No./lb
1/2	.072	.057	531
1/2	.083	.060	399
5/8	.072	.057	461
5/8	.083	.060	345
3/4	.072	.057	406
3/4	.083	.060	305
7/8	.072	.057	364
7/8	.083	.060	273

1" Wide Hoop Staples (Inside Measurement)

Length	Gauge	Flatten To	No./lb
5/8	.083	.060	324
5/8	.109	.083	198
3/4	.083	.060	288
3/4	.109	.083	176
7/8	.083	.060	259
7/8	.109	.083	158
1"	.083	.060	236
1"	.109	.083	143

1-1/4" Wide Hoop Staples (Inside Measurement)

Length	Gauge	Flatten To	No./lb
3/4	.083	.060	179
3/4	.109	.083	120
1"	.083	.060	216
1"	.109	.083	130

3.8 Workmanship. Articles shall be true to shape, well finished, reasonably free from imperfections, clean, and free from corrosion.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other commercial laboratory acceptable to the Government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot. All nails or staples of the same type, class, and size, presented at one time shall be considered a lot for purposes of acceptance, inspections, and tests.

4.3 Sampling.

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4.3.1 *For Inspection.* A random sample of nails or staples shall be taken from each inspection lot offered for Government Inspection of visual and dimensional characteristics, with lot acceptance based on the following sampling inspection requirements.

TABLE XXXI.—Sampling for visual dimensional inspection

Lot Size—number of kegs, pkgs. or containers in inspection lot	Sample Size—number of kegs, pkgs. or containers in sample	Sample Size—number of nails, staples, etc.	Acceptance number of defectives
2 to 15	2	14	1
16 to 40	3	24	3
41 to 110	5	35	5
111 to 300	7	49	6
301 to 500	10	70	9
501 to 1300	15	120	12
1301 to 3200	25	150	17
3201 to 8000	35	215	24
8001 and over	50	300	32

4.3.2 *For lot acceptance tests.* A random sample of nails or staples shall be taken from each inspection lot in accordance with the following table and shall be subjected to the tests specified in 4.5.

TABLE XXXII.—Sampling for physical tests

Lot Size—number of kegs, pkgs. or containers in inspection lot	Sample Size—number of kegs, pkgs. or containers from which sample nails shall be drawn	Sample Size—number of nails, staples, etc. for each applicable test	Acceptance number of defectives in all tests
2 to 15	2	4	0
16 to 40	3	6	0
41 to 110	5	10	0
111 to 300	7	14	1
301 to 500	10	20	1
501 to 1300	15	30	2
1301 to 3200	25	50	3
3201 to 8000	35	70	5
8001 and over	50	100	6

4.3.3 *Additional sampling.*

4.3.3.1 *Copper-clad staples.* If the contract or order includes copper-clad staples, the producer shall furnish with the staples five wire specimens, each not less than 12 inches long, certified to be representative of the wire used in the production of the staples, for the tests specified in 4.6.4 and 4.6.5.

4.3.3.2 *Aluminum nails.* If the contract or order includes aluminum nails, the producer shall furnish with the nails, five wire specimens, each not less than 12 inches long, certified to be representative of the wire used in the production of the nails, for the test specified in 4.6.6.

4.4 Visual and dimensional inspection. Each of the sample nails or staples selected in accordance with table XXXI shall be visually and dimensionally inspected to verify compliance with the specification requirements as to size, finish, workmanship and material. Nonconformance with respect to length of shank or diameter of wire shall be considered major defects; corrosion, faulty shape, poor finish, improper color, and non-uniformity of coating, if applicable, shall be considered minor defects.

4.5 Lot acceptance tests.

4.5.1 *Testing of sample nails and staples for lot acceptance.* Each of the sample nails or staples selected in accordance with table XXXII shall be tested in accordance with paragraph 4.6 to verify compliance with the requirements of this specification. If the number of non-conforming nails or staples in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.6 Tests. Nails and staples shall be given such tests as may be necessary to determine conformance with the requirements of this specification.

4.6.1 *Tests for ductility or hardness.* Articles tested for ductility or hardness shall be bent by any suitable means available.

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4.6.2 *Tests of cement-coated nails.* Tests for immediate holding power (within 24 hours after driving) of cement-coated nails shall be made by comparison with either bright tumbled nails before coating or nails stripped of cement coating (see 6.6). An equal number of cement-coated and uncoated specimens shall be driven into sound, dry, western yellow pine, hemlock, fir or larch, perpendicular to the grain with heads protruding 1/4 inch. The procedure specified in ASTM-D-1037, paragraphs 43-44, (Nail Withdrawal Resistance) shall be followed. Initial withdrawal resistance shall be expressed in percentages of the average resistance of the cement-coated nail above that of the uncoated for each group of identical specimens tested. Thus, if the average resistance of the uncoated is 100 and that of the cement-coated is 160, the increased holding capacity is 60 percent.

4.6.3 *Analysis for aluminum and copper.* Where required by the invitation, contract, or purchase order, a report shall be furnished showing the composition of the aluminum or the purity of the copper used for the production of the aluminum and copper articles included in the order. Tests may be made by the manufacturer of the articles, by the producer of the material from which the articles are made, or by an acceptable commercial laboratory, and may employ any standard method usually used for such determinations. Reports thus furnished shall be certified by the supplier as representing the materials used in the production of articles included in the order.

4.6.4 *Measurement of thickness of copper cladding.* Measurements of thickness of copper cladding shall be made either by (*a*) accurately measuring the diameter of the wire before and after removing the coating by any suitable means, or (*b*) cutting off the wire, grinding smooth and etching the exposed cross section to differentiate between core and coating, and measuring under suitable magnification, or (*c*) using electrical indicating instruments of suitable accuracy.

4.6.5 *Weight of copper cladding.* The determination of the percentage of copper shall be made by chemical analysis of a representative sample of the wire, or by chemically stripping the copper and accurately weighing the sample of wire before and after stripping.

4.6.6 *Tensile strength of aluminum.* Samples of aluminum wire selected in accordance with 4.3.3.2 shall be subjected to a tensile strength test to verify conformance with 3.1.5.

4.6.7 *Zinc-coating.* Sample nails selected in accordance with 4.3 shall be tested to determine conformance with 3.2.1.

4.6.7.1 *Amount of zinc-coating.* The heads and points shall be cut from the samples and the amount of zinc coating on the remaining lengths shall be determined in accordance with method 512.1 of Fed. Test Method Std. No. 151.

4.7 **Inspection of preparation for delivery.** The packaging, packing and marking of items shall be inspected to determine conformance to the applicable requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

(For civil agency procurement the definitions and application of the various levels of packaging and packing are defined in Fed. Std. No. 102.)

5.1 Packaging.

5.1.1 *Level A.*

5.1.1.1 *Poultry-netting staples and common brads.*

5.1.1.1.1 *Unit packaging.* Like items up to and including 1/2 inch in length shall be

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packaged in 1/2-pound quantities. Lengths over 1/2 inch shall be packaged in 1-pound quantities. The containers shall be either commercial cartons, or folding cartons, metal-stayed or set-up boxes conforming to PPP-B-566, PPP-B-665, and PPP-B-676, respectfully.

5.1.1.1.2 *Intermediate packaging.* The packages of like items shall be intermediate packaged in fiberboard boxes conforming to PPP-B-636, type and class at the option of the supplier. The boxes shall be closed and sealed in conformance with the appendix to the box specification.

5.1.1.2 *Nails and fence staples.* Nails and fence staples, when packaging is specified, shall be packaged in 5 pound quantities and in a container selected from those referenced in paragraph 5.1.1.1.

5.1.2 *Level C.* The nails, staples, and brads shall be packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity for either immediate use or for controlled humidity (CH) sotrage.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Poultry-netting staples and common brads.* The intermediate packaged items of like description shall be packed in overseas type or class boxes conforming to any of the following specifications:

PPP-B-576—Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585, class 3—Boxes, Wood, Wirebound.
PPP-B-591—Boxes, Fiberboard, Wood-cleated.
PPP-B-601—Boxes, Wood, Cleated-plywood.
PPP-B-621—Boxes, Wood Nailed and Lock-Corner.
PPP-B-636 V3c or V3s—Box, Fiberboard.

Gross weight of wood and wood-cleated boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the box specification. Box closures and strapping shall conform to the box specification and the appendix thereto. In lieu of steel strapping, fiber boxes may be reinforced with pressure-sensitive filament tape conforming to PPP-T-97, type IV, applied in accordance with the appendix thereto. Boxes shall be provided with case liners conforming to MIL-L-10547, sealed in accordance with the appendix thereto. Case liners are not required when intermediate containers conforming to class 2 of PPP-B-636 are used.

5.2.1.2 *Nails and fence staples.* Nails and fence staples of like description shall be packed in quantities of 50 pounds in one of the following containers as specified. (See 6.1.) The containers shall be designed to limit the void space within the container to a minimum. Packaged material shall be packed as specified in 5.2.1.1.

5.2.1.2.1 *Slack kegs.* Slack kegs shall conform to NN-K-231. The kegs shall be closed and strapped in conformance with the appendix to the keg specification, using type I, class B, grade 2, flat steel strapping.

5.2.1.2.2 *Nailed wood boxes.* Nailed wood boxes shall conform to PPP-B-621, style 2. The boxes shall be closed and strapped in conformance with the appendix to the box specification, using type I, class B, grade 2, flat steel strapping.

5.2.1.2.3 *Wood cleated-plywood boxes.* Wood cleated-plywood boxes shall conform to PPP-B-601 (Overseas Type). The boxes shall be closed and strapped in conformance with the appendix to the box specification, using type I, class B, grade 2, flat steel strapping.

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5.2.1.2.4 *Steel pails.* Steel pails shall conform to PPP-P-704, type II. The body and bottom of the pails shall be fabricated from 24 gage steel, and the top from 24 gage steel. Covers shall be 16 lug type. The pails shall be closed in conformance with the appendix to the container specification.

5.2.1.2.5 *Heavy duty fiberboard boxes.* Heavy duty fiberboard boxes shall conform to PPP-B-636, type II. The boxes shall be closed and strapped in conformance with the appendix to the box specification.

5.2.1.2.6 *Fiber drums.* Fiber drums shall conform to PPP-D-723, type III, grade A. The drums shall be closed in conformance with the drum specification.

5.2.2 *Level B.*

5.2.2.1 *Poultry-netting staples and common brads.* The packaged material shall be packed as specified in 5.2.1.1, except that boxes shall be domestic type or class. Case liners will not be required. The gross weight of fiberboard boxes shall not exceed 90 pounds.

5.2.2.2 *Nails and fence staples.* Unpackaged nails and fence staples of like description shall be packed in quantities of 100 pounds. The containers shall be heavy-duty fiberboard boxes conforming to PPP-B-636 type I, fiber drums conforming to PPP-D-723, type I, grade A, slack Kegs conforming to NN-K-231, nailed wood boxes conforming to PPP-B-621, or wood cleated-plywood boxes conforming to PPP-B-601 (domestic type). Packaged material shall be packaged as specified in 5.2.2.1. When specified in the contract or order (see 6.1), unpackaged nails shall be packed in 50 pound quantities in fiberboard boxes, conforming to the special requirements of PPP-B-636, Class 1.

5.2.3 *Level C.*

5.2.3.1 *Packaged nails, staples, and brads.* Packaged nails, staples, and brads shall be packed in containers to afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. This level as a minimum shall conform to applicable carrier rules and regulations.

5.2.3.2 *Unpackaged nails, staples, and brads.* Unpackaged nails, staples, and brads shall be packed in quantities as specified in containers conforming to PPP-B-636, type I, or shall be packed to afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. This level as a minimum shall conform to applicable carrier rules and regulations.

5.3 Marking.

5.3.1 *Civil agencies.* All shipments for civil agencies shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military.* In addition to any special marking required by the contract or order, all intermediate and shipping containers for any military agency or service shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Ordering data.** Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type required (See 1.2.1).
(*c*) Size (or length in inches) (See 1.2.2 and 3.4).
(*d*) Finish required (See 1.2.3).
(*e*) Material (if required).
(*f*) Shank design required if other than as specified herein (see 1.2.4).
(*g*) Administrative provisions for inspection records (See 4.1).
(*h*) Level of packaging and level of packing required (See 5.1 and 5.2).
(*i*) Container required for packing (See 5.2.1.2.).

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6.1.1 For nails (small) and tacks, cut, see Federal Specification FF-N-103.

6.2 The steel wire gauge numbers which are equivalent to the decimal diameters used in this specification are as below:

6.3 Illustrations are intended to show the general design only and are not intended to preclude compliance with specification requirements.

6.4 It is believed that this specification adequately describes the characteristics necessary to secure the desired articles, and that normally no samples will be necessary, prior to award, to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.5 Federal specifications do not include all types, sizes, materials, finishes, and other descriptions of the items indicated by the titles of the specifications, but are intended to cover those types, sizes, and so forth, which are most suitable for Federal Government requirements.

6.6 **Coating removal.** The coating on nails may be removed by using acetone or a varnish remover conforming to Federal Specification TT-R-251 (Remover paint and varnish, organic solvent type). The nail should be soaked in the acetone for not less than 1/2 hour or in the varnish remover for not less than 1/4 hour. After soaking the nail should be cleaned with a cloth to remove the coating and varnish remover that remains.

6.7 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Nails, Steel not otherwise indexed by name, plain, galvanized, japanned or tinned or coated with brass, bronze, cadmium, cement, or copper.

Carload minimum weight 40,000 pounds.

Motor:

Nails, steel, not otherwise indexed, plain, galvanized, japanned or tinned, or coated with brass, bronze, cadmium, cement or copper.

Motor volume minimum weight 40,000 pounds.

Rail:

Nails, Aluminum.

Carload minimum weight 30,000 pounds.

Diameter, inch	Gage number	Diameter, inch	Gage number	Diameter, inch	Gage number
0.2625	2	0.131	10-1/4	0.072	15
.2437	3	.128	10-1/2	.067	15-1/2
.2253	4	.1205	11	.0625	16
.207	5	.113	11-1/2	.058	16-1/2
.192	6	.1055	12	.054	17
.177	7	.099	12-1/2	.0475	18
.162	8	.0915	13	.0410	19
.1483	9	.086	13-1/2	.0348	20
.142	9-1/2	.080	14	—	—
.135	10	.076	14-1/2	—	—

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Motor:

Nails, Aluminum.
Truckload minimum weight (W) 30.2 pounds, (W) subject to Rule 115, National Motor Freight Classification.

Rail:

Nails, copper, not plated.
Carload minimum weight 30,000 pounds.

Motor:

Nails, copper not plated.
Motor volume minimum weight 30,000 pounds.

Rail:

Staples, Aluminum.
Carload minimum weight 30,000 pounds.

Motor:

Staples, Aluminum.
Truckload minimum weight (W) 30.2 pounds, (W) subject to Rule 115, National Motor Freight Classification.

Rail:

Staples, Bronze.
Carload minimum weight 30,000 pounds.

Motor:

Staples, bronze.
Motor volume minimum weight 30,000 pounds.

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